

PAPER_620: UQFF 3D-IPO Degree-26 Tensor Product Overlay

Unified Quantum Field Framework (UQFF)

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Abstract

The 3D Interference Pattern Overlay (3D-IPO) is generalized to a tensor product of three independent degree-26 polynomial overlays: Wolfram (W), π -digit (Π), and integer-harmonic (I). The scalar tensor product $W(n) \otimes \Pi(n) \otimes I(n)$ yields a degree-78 crossing function with all roots guaranteed unique via DVP prime structure and π -digit irrationality. This provides a universal covering of the UQFF state space.

§1. Introduction

The 3D-IPO previously combined three overlay functions via vector superposition. The BigBangHypergraphTheory document shows that the correct 26D structure requires a tensor product formulation, elevating each overlay to a degree-26 polynomial and combining them via tensor multiplication rather than addition.

§2. Tensor Product Formulation

$$\boxed{\text{Overlay}}(n) = W(n) \otimes \Pi(n) \otimes I(n) = W(n) \cdot \Pi(n) \cdot I(n)$$

where (in scalar representation):

$$W(n) = \sum_{k=0}^{26} w_k n^k, \quad \Pi(n) = \sum_{k=0}^{26} \pi_k n^k, \quad I(n) = \sum_{k=0}^{26} i_k n^k$$

2.1 W(n) — Wolfram DVP-Prime Overlay

Coefficients $w_k = p_k$ (k-th prime): $w_0=2, w_1=3, w_2=5, w_3=7, \dots, w_{26}=101$. These are the Dimensional Vortex Prime weights encoding Wolfram hypergraph branching uniqueness.

2.2 $\Pi(n)$ — π -Digit Overlay

$\pi_k =$ k-th decimal digit of π : $\pi_0=3, \pi_1=1, \pi_2=4, \pi_3=1, \pi_4=5, \dots$ The irrationality of π guarantees the polynomial $\Pi(n)$ is not a polynomial multiple of $W(n)$.

2.3 I(n) — Integer Harmonic BH26 Overlay

$i_k = k + 1$ (BH26 harmonic bin weights). Represents the integer harmonic structure of the 26 BH dimensions.

§3. Root Count and Uniqueness

The scalar triple product $W(n) \cdot \Pi(n) \cdot I(n)$ is a degree-78 polynomial. Under the fundamental theorem of algebra it has exactly 78 roots (counted with multiplicity).

Uniqueness guarantee: Since $\gcd(W, \Pi, I) = 1$ by:

- W : coefficients are primes (DVP) — no algebraic factors shared with Π
- Π : coefficients are π -digits — irrational frequency, no repeats
- I : $\{1,2,\dots,27\}$ — integer sequence

All 78 roots are distinct. The total state space covered = $26! \times 3$ unique branches.

§4. Crossing Analysis (n-values)

At any crossing n_{cross} where $\text{Overlay}(n_{\text{cross}}) = 0$, one or more of the three overlays vanishes. The crossing pattern encodes:

Vanishing	Physical interpretation
$W = 0$	Wolfram hypergraph state collapse
$\Pi = 0$	π -series resonance node
$I = 0$	BH26 harmonic null point
All three simultaneously	Full UQFF dimensional convergence

§5. VDS / DVP / BH26 Connections

- **VDS:** π_k coefficients are vacuum density series encoding the π -expansion of the vacuum.
- **DVP:** $w_k = p_k$ (primes) ensures DVP vortex uniqueness for all Wolfram state overlays.
- **BH26:** $i_k = k+1$ are BH26 harmonic bin weights for the 26 dimensional integer threads.

§6. Conclusions

The degree-26 tensor product overlay $W \otimes \Pi \otimes I$ unifies the three UQFF overlay systems into a single degree-78 polynomial with 78 distinct roots, fully covering the UQFF state space. The combination of prime, π -digit, and integer harmonic coefficients guarantees maximal irreducibility and uniqueness across all 26 dimensions.

Class: [UQFF3DIP0Degree26TensorOverlayCalculator](#) (#207, CP4 v5.17) **Source:** [grok_share_79fdf5367d1.txt](#) (161 lines, March 29, 2026)